Learning airline route preferences from flight-plan revisions

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Abstract

Every revision of a flight plan records a choice that an airline actually made, namely the route it dropped and the route it filed instead. Eighteen months of European traffic contain 1.48 million such pairs, a continuously refreshed sample of revealed route preference, and a stronger label than the generated alternatives of earlier work, which the airline may never have seen. This paper fits a pairwise ranking model on 1 134 000 of these pairs, expressing each route's indicators relative to the other route's, and identifies the route the airline chose in 68.1 % of held-out decisions. Simple cost rules do strictly worse than chance: preferring the route with less planned fuel scores 45.0 %. The model knows its own uncertainty, reaching 94.9 % on its most confident fifth of cases, so a proposal channel can operate at a precision fixed in advance. Simulated on a held-out quarter at 80 % target precision, such a channel surfaces 20.1 kilotonnes (kt) of planned fuel per quarter that the airlines' own later filings confirm, of order 220 to 250 kt of CO_2 per year, four fifths of it flown by 28 of the 882 operators. Trained on one year and watched month by month over the next half year, the model's proposal precision moves by under two points while its coverage falls by about a fifth: it ages by proposing less rather than by erring more. While there is still room for improvement, particularly on flights that never revise their plans, these figures indicate that airline route acceptance is learnable from operational records alone.

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1. Introduction

On a spring morning in 2026, an Airbus A320 was filed from Amsterdam to Barcelona on a route carrying 102 minutes of attributed air traffic flow management (ATFM) delay, and the operator subsequently re-filed. The new route shed that delay completely, at a cost of 39 minutes of flight time and 1 397 kg of extra planned fuel. Both routes are held in the network's records and so, implicitly, is the operator's judgement that the delay was worth more than the 1.4 tonnes of fuel that avoiding it consumed.

That judgement is the quantity this paper models. A flight plan is a negotiated object: an airline files, the network responds with regulations and delay, and the airline may file again. Each revision records the route replaced and the route that replaced it, and therefore a preference that the operator expressed with its own filing system rather than in answer to a survey.

The operational interest is concrete, and the network-level context is reported annually by the Performance Review Commission (Performance Review Commission, 2026): airlines frequently revise onto routes that *cost* fuel in order to escape delay, while fuel-cheaper alternatives for other flights are not identified in time to be filed. The difficulty is not generating candidate routes; it is knowing which candidate is acceptable to this particular operator, on this day, with this rotation to protect, and knowing it inside the planning window rather than under tactical time pressure. Needless to say, proposing routes that airlines reject is worse than proposing nothing at all, because a channel that is ignored is a channel that is eventually switched off.

A preliminary version of this work was presented at the SESAR Innovation Days 2026. The present paper extends it with the material a feasibility claim actually rests on: the composition of the archive and the behaviour it records, the evidence that simple cost rules fail on this task, the strata in which the model earns and loses its accuracy, the sensitivity of every fuel figure to the bias correction, and a retraining-horizon experiment in which the model is trained on one year and watched, month by month, over the following half year.

The contribution is fourfold. First, this paper shows that airline route acceptance is learnable at network scale from revision events alone, and that the confidence of the resulting model is calibrated well enough to support a chosen operating point rather than a single accuracy figure. Secondly, it establishes that the learned preference actively contradicts planning cost: every rule of the form "prefer the cheaper route" scores below chance on held-out revisions, which is the cleanest evidence available that filing behaviour carries structure a cost model cannot represent. Thirdly, it quantifies what a proposal channel built upon such a model could address, using the airlines' own subsequent filings as ground truth, together with how that value concentrates across operators and how sensitive it is to the design decisions. Fourthly, it measures how quickly the model ages, which is the question a deployment plan needs answered and the one a single train–test split cannot address.

The remainder of this manuscript is organised as follows. Section 2 explains why a pair of routes drawn from a revision constitutes a preference observation, and Section 3 positions the work within the literature on rerouting, route choice and learning from implicit feedback. Section 4 describes the archive and the construction of the training pairs, and Section 5 is devoted to the design problem that dominates this construction, namely leakage. Section 6 describes the model, its training and its calibration. Section 7 reports predictive performance and its structure, Section 8 the proposal-channel simulation, and Section 9 the retraining-horizon experiment. Section 10 collects the threats to validity, and Section 11 concludes.

2. Background

This section describes fundamental concepts that are not a contribution of this work, but that are needed to follow the sections that come after it. Thus, a reader who is already familiar with the European filing process and with discrete choice may move on to Section 3.

2.1. Flight-plan revisions as preference data

In the European network, a filed flight plan may be revised before departure. When the revision changes the horizontal route, the system records a route change event containing the flight-plan message that was replaced and the message that replaced it. Each event therefore yields two routes for the same flight, the same day and the same operator, exactly one of which the airline chose.

When capacity is constrained, the network applies regulations (temporary capacity restrictions declared on volumes of airspace) and assigns each crossing flight a delay; this is the attributed ATFM delay used throughout the paper. An operator facing such a delay has a choice that is genuinely economic: accept it, or file a route that avoids the constrained volume at the cost of distance, fuel and en-route charges. Neither answer is universally correct. For instance, a flight with a tight aircraft rotation may find forty minutes of delay far more expensive than a tonne of fuel, whereas the same delay on the last leg of the day may well be cheaper than the detour. This is why the decision is worth learning rather than deriving: the trade depends on operator-specific and flight-specific costs that the network does not observe, and that airlines weigh against published reference values for the cost of delay (Cook and Tanner, 2015).

The archive reflects this trade directly. Slightly under a third of the revision pairs involve any attributed delay at all, and the newly filed route is more often longer, slower and more fuel-intensive than the one it replaced. Consequently, a proposal channel is not looking for the common case, but rather for the minority of flights on which a genuinely cheaper alternative exists and would be accepted.

2.2. Relation to discrete choice

Readers of this journal will recognise the setting as a route-choice problem, a family with a long history in travel-demand modelling (McFadden, 1974; Ben-Akiva and Lerman, 1985; Train, 2009). The observation unit here is the choice between exactly two alternatives that the decision maker itself constructed, which sidesteps the choice-set generation problem that route-choice models must otherwise solve, and the estimator is a gradient-boosted ranker rather than a parametric logit. Roughly speaking, the model of this paper is a conditional logit whose utility function is learned nonparametrically: the training objective penalises the model whenever the abandoned route scores at least as high as the chosen one, exactly as a binary logit on utility differences would, but the score is an arbitrary function of the features rather than a linear index. What is given up is the economic interpretability of coefficients; what is gained is the ability to absorb the waypoint sequence itself, which Section 7.7 shows carries more of the decision than any cost indicator. The economic tradition supplies the framing that matters here, namely revealed rather than stated preference (Samuelson, 1938): every label in this paper is something an airline did, not something it said it would do.

2.3. Why ranking rather than classification

The operational question is comparative: given two routes for one flight, which of them does the airline prefer? Absolute quantities are close to meaningless across the network, since a 2 000 kg fuel figure means one thing on a short-haul leg and quite another on a long-haul sector. The key performance indicators of each route are therefore expressed relative to those of the other route in the pair, by subtracting the pair minimum from each: what reaches the model is not 2 000 kg but the 300 kg by which one route exceeds the other, which carries the same meaning whether the sector is 400 or 4 000 km. Section 7.4 shows that this relative encoding is the single most important design choice in the system.

3. Related work

Three threads of literature meet in this paper: rerouting in air traffic flow management, route choice as an economic decision, and learning preferences from logged behaviour.

Work on rerouting in air traffic flow management has largely approached the problem from the network side. Over two decades ago, rerouting was formulated as a dynamic network flow problem (Bertsimas and Patterson, 2000), and a stochastic extension later rerouted flights as capacity evolved (Mukherjee and Hansen, 2009). Both optimise an objective defined by the network and assume, in effect, that the resulting assignment is executed, so that the willingness of the airline to fly it is not modelled. The value of airline collaboration in flow management has been examined (Castelli et al., 2012), which motivates the proposal channel simulated herein but does not address which proposals would be taken up. The closest work in this thread incorporates airline preferences into collaborative rerouting decisions and shows that accounting for them changes which reroutes are selected (Murça, 2018). Those preferences, however, are inputs to an optimisation, whereas this paper learns the preference function itself, at network scale, from what airlines did rather than from what they stated. Beyond the research literature, airlines already express their priorities through the user-driven prioritisation process (Pilon et al., 2021), a legacy and trusted mechanism that a preference model of this kind would complement rather than replace.

Almost concurrently with the network-side work, route choice in Europe was studied as a trade between fuel and en-route charges (Delgado, 2015), which establishes that the choice is economic and therefore learnable, but

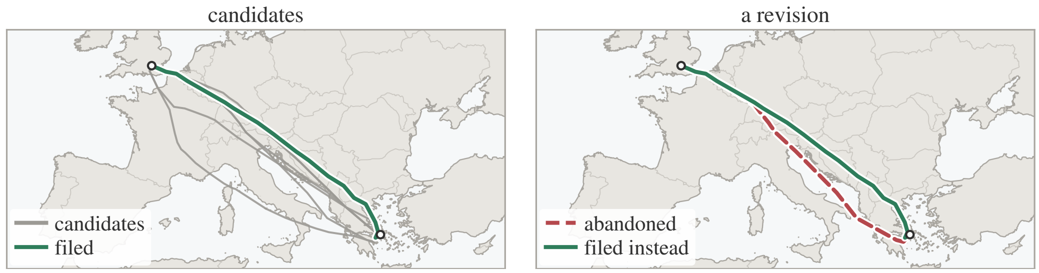

Figure 1: Two ways to build a training pair, on one city pair. Left: the filed route (solid green) is ranked above candidates from a route generator (grey), which the airspace user may never have evaluated; those drawn here are other routes observed on the same city pair. Right: a revision supplies two routes that the same operator filed for the same flight hours apart, one of which it abandoned. Only the right-hand pair records a preference the airline actually expressed.

which uses no label in which the operator rejected an alternative it had itself filed. The discrete-choice tradition discussed in Section 2.2 supplies the theoretical frame for such labels; what it has lacked in this domain is a data source in which both alternatives were genuinely on the decision-maker's table. Revisions are exactly that source.

The third thread is methodological. Learning a ranking function from pairs is standard (Burges et al., 2005; Liu, 2009), and learning preferences from logged behaviour rather than from elicited judgements was established in information retrieval, where clicks on search results were shown to yield reliable relative (though not absolute) relevance labels (Joachims, 2002). The revision archive is the aviation analogue of a click log: nobody was asked anything, and each record is trustworthy only as a comparison between the alternatives that were actually on the table. The same literature also supplies the caution that this paper inherits, namely that logged data reflects the behaviour of the system that logged it, a point returned to in Section 10.

The direct predecessor of this work applies learning to rank to flight routes for a different purpose: predicting the flight plans that have not yet been filed, so that traffic demand can be estimated earlier than the filing horizon allows (Dalmau et al., 2024). That work established the framing adopted herein, namely a gradient-boosted ranker that scores routes by their indicators in context, and it reported the very difficulty that motivates the present paper. Its training pairs take the filed plan as preferred and a set of generated proposals (up to eleven route options produced by the Network

Manager) as less preferred, and on the pre-tactical set none of the latter need match the filed route at all. Ranking the filed plan above such candidates may therefore require only that the model recognise which route looks like a filed plan, which is a different and much easier skill than knowing which route an operator would prefer, and accuracy alone cannot distinguish the two. For instance, a generated candidate may cross airspace that the operator never uses, or follow a track that its own flight-planning system would discard on company policy; telling such a route from a filed plan calls for no knowledge of preference whatsoever. No dataset offers both label sources for the same flights, so this is an argument about how the labels are built rather than a measured comparison.

Revisions remove that difficulty (Fig. 1). Both routes were filed by the same operator, for the same flight, on the same day, so the negative is not a plausible alternative constructed after the fact but the route that the airline had committed to and then abandoned. The question also narrows: the model is not asked what an airline will file, but which of two specific routes it prefers, which is a strictly easier question and precisely the one a proposal channel needs answered. The price to be paid is that revisions are not a random sample of flights, but rather flights that changed, which introduces a bias of its own. Rather than assuming that bias away, Section 7.6 measures it.

4. Data and pair construction

Figure 2 shows the procedure end to end, from the archive to a proposal. This section covers its first two stages, the construction of the pairs; the model and the calibration follow in Section 6.

4.1. The archive

Eighteen consecutive months of European network data, from January 2025 to June 2026, are used. After retaining only those revisions that change the horizontal route, and only those pairs with complete key performance indicators, the archive contains 1.48 million labelled pairs. Each route is described by twenty-three features, summarised in Table 1.

All the data used is held by the network for operational purposes and was processed under existing data-handling arrangements; accordingly, operator identity enters the model as a categorical feature only, and no operator, flight

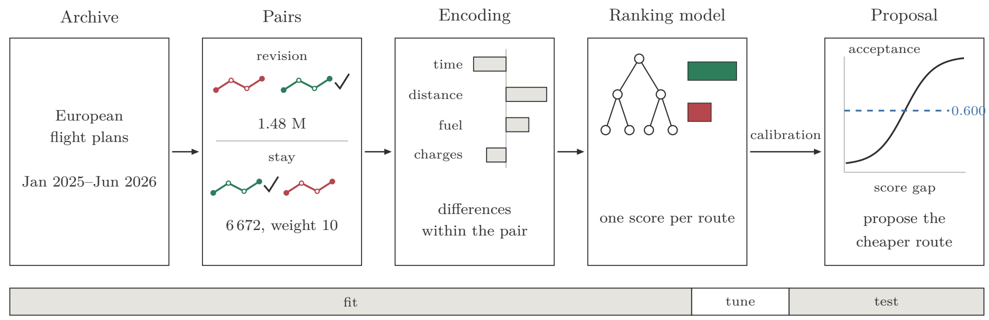

Figure 2: How a filing archive becomes a proposal. Each revision yields two routes for the same flight, exactly one of which the airline chose (tick); stay pairs (flights that kept their route, Section 4.3) invert that construction and enter at a group weight of 10. The score gap becomes an acceptance probability, compared with a threshold fixed in advance.

Table 1: What the model sees for each route; some rows summarise several features (twenty-three in all, seven regulation-derived). Indicators enter as differences within the pair; the rest are identical on both routes and act as context.

group	content
indicators	flight time, distance, planned fuel, route charges
regulation	attributed delay (own, inbound, outbound), their count, reason, protected volume
rotation	inbound and outbound connection times and states
context	operator, aircraft type, city pair, hour, weekday, month, time to departure
route	the ordered waypoint sequence, as text

or airspace volume is identified anywhere in this paper, so that no finding can be read as an assessment of any operator or airspace.

4.2. What the archive records

A journal reader deserves to know what these 1.48 million decisions actually look like before any model touches them, and Table 2 therefore breaks the archive down by the regulation context in which each revision was made.

Three observations follow. First, seven revisions in ten involve no attributed delay on either route: most of the archive records ordinary flight-plan housekeeping rather than escapes from regulations, and the median fuel difference within those pairs is a single kilogramme. Secondly, when a revision does escape a regulation, it pays for the escape: only 28.1 % of escape

Table 2: The revision archive by decision context: unregulated pairs carry no delay on either route, escapes carry delay on the abandoned route only, the next two contexts carry delay on both, and in the final context only the newly filed route carries delay. The median fuel difference is between the two routes of a pair; the final column is the share of pairs in which the newly filed route burns less planned fuel than the one it replaced.

context	pairs	share [%]	median Δfuel [kg]	new route saves fuel [%]
unregulated	1 047 228	70.6	1	46.2
escaped	117 594	7.9	101	28.1
reduced delay	155 326	10.5	65	31.6
kept or raised delay	141 648	9.6	0	47.4
took on delay	21 337	1.4	-5	51.8

revisions move to a route that burns less fuel, and the median move costs 101 kg and 2.1 minutes. Thirdly, the only context in which the share of fuel-saving moves exceeds one half is the smallest one, the 1.4 % of revisions that took on delay, and it is the mirror image of the escape: a flight that accepts delay tends to be paid in fuel for it. Everywhere else the share stays below one half, which already suggests that fuel is not what filing behaviour optimises, an observation that Section 7.2 sharpens into a measured result. Figure 3 shows two revisions drawn from the archive, an escape and a fuel saver.

4.3. Stay pairs: the missing half of the question

Every revision pair is, by construction, a flight that *did* change route, and that creates a trap. A model trained only on such events learns that change is what happens, and recommends it far too eagerly against traffic that mostly stays: an uncorrected model prefers the alternative route in 65.2 % of the cases in which the flight demonstrably kept its own. Put another way, the archive answers one question only, namely: when an airline changed its route, which route did it move to? It cannot answer the complementary question, because a flight that kept its route generates no record of the alternative that it declined.

A second, much smaller dataset was therefore constructed, in which *keeping* the route is the revealed choice. Let us consider two flights on the same day, with the same origin, destination and aircraft type, that flew different routes and neither of which revised its plan, and let us suppose that one of

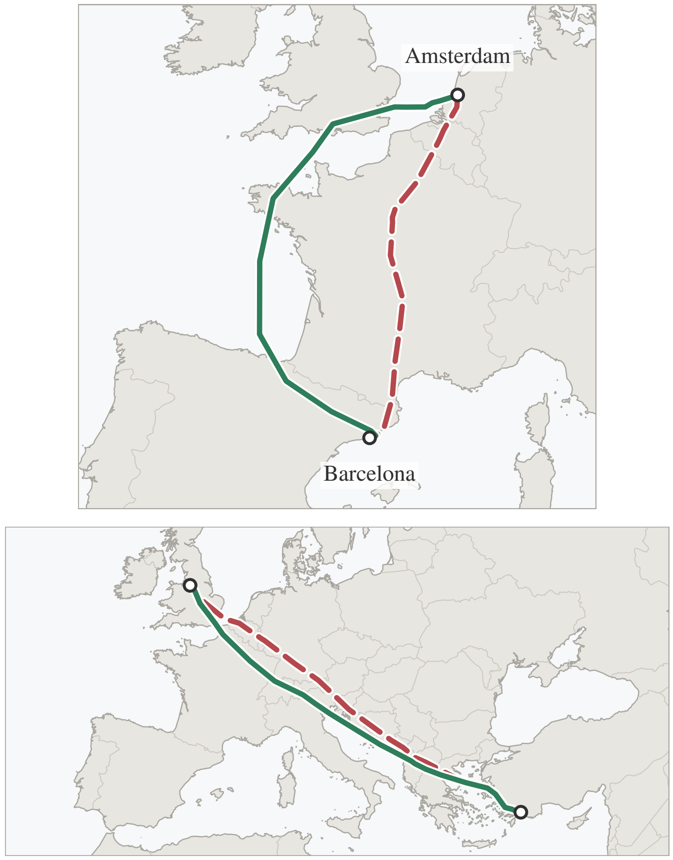

Figure 3: Two revisions from the archive, in the visual language used throughout this paper: the abandoned route is dashed red, its replacement solid green. Top: an escape, in which the operator paid fuel and time to leave a regulated volume. Bottom: a fuel saver, in which the replacement was cheaper outright. The first kind dominates the regulated part of the archive; the second is what a proposal channel exists to surface earlier.

them was caught by a regulation while the other was not. The regulated flight had both a reason to move and a demonstrated alternative, namely the route that its twin was flying that same day, and yet it stayed; its route is accordingly labelled as preferred, and that of its twin as the alternative it declined. The conditions are restrictive and the yield is correspondingly low: 6 672 such *stay pairs*, of which 1 093 fall in the test months, against 1.48 million revision pairs. The two are used together, with stay pairs entering the fitting set at an increased group weight (the pair's weight in the ranking loss).

The stay pairs also carry evidence of their own. The flight that stayed faced a median peak delay (the largest delay attributed to it at any point before departure) of 31 minutes, and in 47.5 % of the pairs the route it kept burns *more* planned fuel than the alternative its twin was flying (the median difference is 3 kg in the kept route's favour, i.e. essentially nil). Operators therefore sit on delay even when a demonstrated, comparably priced alternative exists, which is revealed preference for stability itself, and precisely the behaviour an over-eager proposal channel would violate.

One property of this construction should be stated here rather than left to the results. Because the regulated flight is always the one whose route was kept, regulation status identifies the labelled answer within the stay dataset. A rule as crude as "prefer the route of the flight that was regulated" is therefore right on every stay pair by construction, without examining a route at all, and a model with access to the delay features can score well on those pairs for the same empty reason. Accordingly, when the change-bias metric is reported, the seven regulation-derived features are masked at scoring time, which forces the decision onto route geometry and indicators. Training still sees them, so the masked figure measures what transferred to the route features rather than what was learned in their absence. It should be noted that this is a repair applied after the fact rather than a clean design, and Section 7.6 quantifies how much it limits what can be concluded.

4.4. Splits

The evaluation is strictly out of time, in the sense that the model is fitted on January 2025 through February 2026, tuned on March 2026, and evaluated once on April to June 2026 (281 720 pairs), the held-out quarter cited throughout. Every threshold, calibrator and stopping decision is taken on the tuning month alone. Flights appearing in the tuning or evaluation months are removed from the fitting set, and eleven automated checks verify split disjointness, feature construction and the pair invariants. A second split, used only in Section 9, moves the entire window back one year so that the model can be watched for six months beyond its training cutoff.

5. Leakage, and the encoding that prevents it

Treating a revision as a labelled preference pair requires care, because the two routes are separated in time as well as in geography: the replacement is by construction the later message, and any feature that drifts with time can therefore identify the answer without saying anything at all about the quality of the route. This section is devoted to that problem, because on this archive it is not a technicality: a model built without attention to it would report excellent figures while knowing nothing whatever about routes.

For instance, the time remaining to departure is necessarily smaller for the route that replaced the other, so a model given that feature in its raw form could answer "the one filed later" and be right on essentially every pair while knowing nothing about routes. This is leakage rather than learning, and it would be invisible in the accuracy figure: the model would score close to perfectly on held-out revisions, for the empty reason that held-out revisions carry the same timestamp asymmetry as the training ones. To remove the shortcut, all flight-level attributes were harmonised across the pair so that only route-level quantities differ, and the filing-time feature was collapsed to the pair minimum, which gives it an identical value on both routes.

A subtler shortcut hides in the absolute indicator values. Let us consider a 400 km sector on which the two routes burn 2 000 and 2 300 kg. A model shown the raw magnitudes learns, inevitably, what 2 000 kg means on this kind of sector, and with it everything that correlates with sector length: which operators fly it, at which hours, with which aircraft. Some of that is legitimate context, but it also hands the model a way to answer from the flight rather than from the comparison, and it generalises poorly to sectors of other lengths. The within-pair encoding subtracts the pair minimum from each indicator, so that what reaches the model is 0 against 300 kg, a pure comparison that carries the same meaning whether the sector is 400 or 4 000 km.

6. Model, training and calibration

This section covers the remaining stages of Fig. 2: the scoring model, the objective it is trained under, and the calibration step that turns its scores into acceptance probabilities. The design follows the predecessor system (Dalmau et al., 2024) except where stated, and the emphasis here is on the two decisions that the results show to matter, namely the within-pair encoding and the stay-pair mixture.

6.1. The ranking objective

The model gives each route a single number, which is not a probability and has no units, and which is only meaningful next to the number given to another route for the same flight. Training pushes those numbers apart in the right direction, in that for every historical pair the model is penalised whenever the abandoned route scores at least as high as the chosen one, and the penalty shrinks as the gap between them grows (Burges et al., 2005). Nothing in the objective asks the model to reproduce a route in isolation, which is what makes the pair the natural unit, and which is also what connects the estimator to the conditional-logit tradition of Section 2.2.

Gradient-boosted decision trees (Prokhorenkova et al., 2017) are used as the scoring function, at a depth of 8, with the learning rate selected automatically and with early stopping on the tuning month. Key performance indicators are normalised within each pair by subtracting the pair minimum. The ordered waypoint sequence is carried as text and encoded as a bag of tokens (i.e., each waypoint identifier is treated as a word and the route as the unordered collection of its words), which lets the model exploit route structure rather than aggregate distance alone. For instance, two routes on the same city pair may differ by under one per cent in distance and in planned fuel, and so be indistinguishable on every indicator, while only one of them crosses a volume whose identifier the model has met in thousands of earlier revisions; the token set carries that fact and the aggregate distance cannot. Stay pairs enter the fitting set with a group weight w, and the weight is swept over the values 10, 30 and 100; the selection of $w = 10$ is discussed with the change-bias results in Section 7.6.

One detail proved consequential, and is worth recording for anyone reproducing this design. When the fitting set is a weighted mixture of revision pairs and stay pairs, the tuning set must carry the same mixture at the same weights, since a tuning set composed only of revision pairs measures a different objective from the one being optimised. Unfortunately, this mismatch caused training to halt at the very first iteration and to return a below-chance model, which is a measurement failure that presents itself as a model failure, and which cost the authors some days of confusion before it was traced.

6.2. From scores to probabilities

Roughly speaking, a score gap of 3.7 tells an operations room nothing at all, and before the model can be used to decide anything its scores have to be turned into statements of the form “about eight times in ten, an airline offered this route will take it”. That conversion is what calibration provides, namely a monotone mapping from score gap to observed acceptance rate, fitted on the tuning month alone (Zadrozny and Elkan, 2002). Three mappings were compared on the tuning month: the raw sigmoid of the score gap, Platt scaling (a two-parameter sigmoid fit) and isotonic regression (a stepwise monotone fit). Isotonic regression was selected on tuning-month log loss, and is what the pipeline uses. Miscalibration is common in this class of model (Guo et al., 2017), but it is worth reporting that on the held-out quarter the raw sigmoid was already well calibrated, at an expected calibration error (the mean gap between stated probability and observed rate) of 0.8 percentage points against 1.7 for isotonic regression, so that the fitted mapping is a safeguard here rather than a large correction. It was selected on the tuning month without inspecting the evaluation months, and that outcome is reported rather than revised after the fact (Gneiting and Raftery, 2007).

Calibration is also what makes a precision target meaningful, since without it one can rank proposals but cannot decide how many of them to make. On the held-out quarter, the stated and observed rates agree to within 1.7 percentage points on average. The proposal policy built on this mapping is described with the channel results in Section 8.

7. Predictive performance and its structure

This section reports how well the model predicts, and then takes the headline apart: which rules fail where the model succeeds, in which strata the accuracy actually lives, which design choices earn their place, and what the errors look like.

7.1. The headline, and the curve behind it

On the held-out quarter the model identifies the airline's chosen route in 68.1 % of 281 720 pairs (95 % interval 67.9–68.3, or 67.7–68.6 under city-pair clustering). Random choice scores 50 %. Reassuringly, the tuning month scores 67.5 %, and there is therefore no sign of overfitting to it.

Accuracy alone understates operational value, because the model is informative about its own uncertainty. Fig. 4 reports accuracy when the model is permitted to abstain on those cases in which the two routes score close together. Let us call the share of flights that the model is allowed to answer its coverage, the remaining flights being left alone. On its most confident fifth of the cases the model is right 94.9 % of the time, and a proposal channel is not obliged to answer every flight: it operates at the left of this curve.

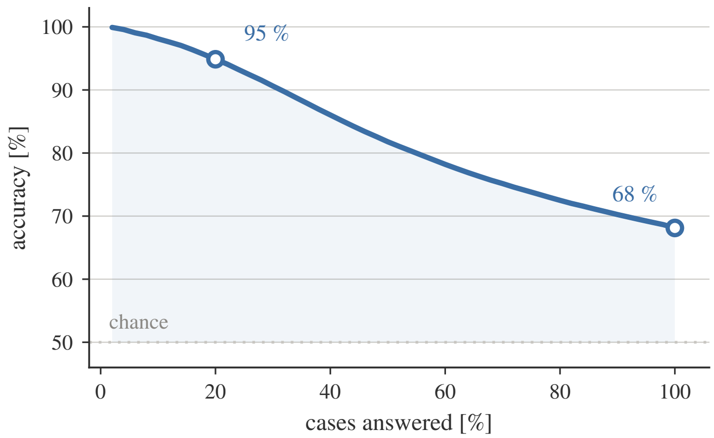

Figure 4: Accuracy against coverage. Moving left, the model answers fewer cases and is right more often.

Table 3: Single-feature rules on the held-out quarter. A rule decides only where the two routes differ on its indicator; the final column is its accuracy on those pairs, and the first accuracy column scores the undecided pairs as a coin toss. Rules preferring the cheaper route score below the 50 % of a coin toss.

rule	accuracy [%]	decides [%]	accuracy where it decides [%]
prefer less planned fuel	45.0	97.9	44.9
prefer less flight time	46.8	96.7	46.7
prefer shorter distance	46.2	81.1	45.3
prefer lower route charges	47.2	53.6	44.8
prefer less attributed delay	53.2	14.6	71.6

7.2. Cost rules score below chance

Table 3 evaluates the single-feature rules a practitioner would try first, each preferring the route that is cheaper on one indicator. Every cost rule scores *below* chance: preferring the route with less planned fuel identifies the airline's choice in 45.0 % of pairs, less flight time in 46.8 %, shorter distance in 46.2 % and lower en-route charges in 47.2 %.

This result deserves a moment, because it is the cleanest statement of why this problem needs learning at all. Filing behaviour does not merely fail to minimise planning cost; it runs against it more often than not, for the structural reason that Table 2 already showed, namely that most revisions buy something (a regulation escape, a slot, operational room) and pay for it in fuel and time. A proposal system built on "offer the cheaper route" would therefore disagree with the airline's own subsequent behaviour on most revised flights. The one partial exception is instructive: preferring the route with less attributed delay is right 71.6 % of the time, but only ever decides on the 14.6 % of pairs where delay differs, and scores 53.2 % overall. Delay avoidance is real, and the model must and does learn it, but it explains only a modest corner of the archive.

7.3. Where the accuracy lives

The 68.1 % headline is an average over strata whose difficulty differs enormously, and Table 4 reports the decomposition. On pairs in which neither route is regulated, a stratum holding somewhat over six pairs in ten of the held-out quarter, the model scores 64.0 %; on pairs in which the old route was regulated and the new one escapes, it scores 88.0 %. Between these poles the accuracy climbs steeply with the delay on the abandoned route: 63.9 % at zero, 70.3 % up to a quarter of an hour, 79.3 % up to half an hour, and about 84 % beyond it, where the climb levels off. The pattern is exactly what an economic reading predicts: the larger the delay, the clearer the incentive, and the easier the decision is to call.

The final block of the table reads strangely at first and repays attention. When none of the three cost indicators favours the newly filed route, so that the airline moved onto a route worse on every recorded cost, the model still identifies the move 79.9 % of the time; when all three favour it, accuracy falls to 53.9 %. In fact, the two results are one result. A move against all the indicators is almost always a move with a strong non-cost reason, typically a regulation, and such reasons are visible to the model; a move that merely tidies up a route the indicators already preferred carries little signal about *when* the operator bothers to make it. The model, in other words, earns its keep precisely where cost logic fails, and adds least where cost logic was already right, which is the correct division of labour for a system intended to complement cost models rather than replace them.

7.4. Which design choices matter

Table 5 reports screening ablations, each configuration removed in turn and the model retrained.

Table 4: Accuracy by stratum on the held-out quarter (Wilson 95 % intervals, the standard interval for a proportion). The final block counts how many of the three cost indicators (fuel, time and charges; distance, the fourth indicator, is not part of this count) favour the newly filed route.

stratum	pairs	accuracy [%]
neither route regulated	178 370	64.0 [63.8, 64.2]
escape (old route only)	30 953	88.0 [87.6, 88.3]
new route only regulated	4 810	58.6 [57.2, 60.0]
both routes regulated	67 587	70.6 [70.2, 70.9]
delay on old route: none	205 205	63.9 [63.7, 64.1]
up to 15 min	19 672	70.3 [69.7, 71.0]
15 to 30 min	18 283	79.3 [78.7, 79.9]
30 to 60 min	21 231	84.5 [84.0, 85.0]
above 60 min	17 329	83.5 [82.9, 84.0]
cost indicators favouring new: 0 of 3	90 961	79.9 [79.7, 80.2]
1 of 3	87 471	70.0 [69.7, 70.3]
2 of 3	78 552	56.8 [56.5, 57.2]
3 of 3	24 736	53.9 [53.3, 54.5]

Expressing the indicators relative to the other route in the pair is decisively the most important choice, since removing it costs nearly ten percentage points: it is what converts four absolute magnitudes into a comparison. The waypoint-text result requires care, and illustrates why screening ablations can mislead. Interestingly, at screening scale the text feature appears worthless, and yet retrained at full scale it is worth +1.4 percentage points (68.2 against 66.8), the reason being that a bag-of-words representation spread over thousands of rare tokens cannot repay its cost in 800 shallow rounds. It is a real but secondary contributor, worth about a tenth of the model's margin over the best single rule of Section 7.2. Yet attribution ranks the same feature first, an apparent tension that Section 7.7 resolves.

7.5. What the errors look like

Fortunately, errors concentrate where the model is already unsure: in 27.7 % of the errors the two routes scored within a quarter of the median margin, and only 5.6 % of the errors sit in the top quarter of the score margin, against the 25 % that would obtain if error were independent of confidence. This is the property that makes confidence filtering work.

Table 5: Screening ablations. Each variant is retrained from scratch at a reduced screening setting (depth 6, 800 rounds) that reaches 60.6 % on revision pairs; values are the change in revision accuracy against that baseline, not headline accuracies. A positive value means the variant scores *higher* on revisions than the adopted configuration does. For the stay-pair correction that is expected: it is the price paid for the change-bias reduction of Section 7.6. Differences are significant under McNemar's test, which asks whether two models' disagreements on the same flights fall asymmetrically rather than by chance, at $p < 0.001$ except the waypoint text at screening scale ($p = 0.58$). The monotone constraint (the score forced never to rise as the number of regulations on a route grows) costs a fifth of a point, which is why none is imposed.

variant	Δ accuracy [pp]
without within-pair KPI normalisation	−10.0
without stay-pair correction	+3.7
with monotone constraint	−0.2
without waypoint text	+0.03
including vertical-only revisions	−2.0

The confident errors are nevertheless instructive, and one example from the held-out quarter is worth following. A business jet on a long-haul flight re-filed off a route carrying 124 minutes of attributed delay onto a route that carried no delay at all, saving 152 kg of fuel and about a minute of flight time; every indicator the model sees favoured the move, and yet the model confidently backed the abandoned route. The flight-level context is identical within a pair by construction, so the preference can only have come from the waypoint sequence itself, presumably a learned structural preference that overrode the indicators; confident errors are therefore not all of the hidden-information kind (decisions driven by facts, like a slot swap or a curfew, visible in no route-level record).

Opportunely, such cases cost less than their share of the error rate would suggest, because a proposal channel would have said nothing at all here: the alternative that it would have offered was the one already filed. The expensive errors are the opposite ones, in which the model confidently proposes a route that the operator declines, and the change-bias measurement reported next is the closest available proxy for their rate.

7.6. Change bias and its partial correction

Measured on the held-out stay pairs, a model trained without correction prefers the alternative route in 65.2 % of cases where the flight kept its own, and when regulation features are masked so that only route understanding remains it scores 47.8 %, below chance. Adding stay pairs at group weight 10 raises the masked figure to 58.1 % (intervals $[55.1, 61.0]$ against $[44.8, 50.7]$, non-overlapping) at a cost of about one point on the revision metric.

The group weight w was selected by a sweep over $\{10, 30, 100\}$, training one model per weight; the sweep's own $w = 10$ run lands within half a point of the adopted model's figures (58.5 against 58.1 on the masked metric), which is the run-to-run variation these numbers carry. Raising w from 10 to 100 improves the unmasked stay figure from 90.3 % to 97.6 %, but the masked figure only from 58.5 % to 59.7 %, while revision accuracy falls from 68.2 % to 66.7 %. Most of what the additional weight buys is therefore the construction's giveaway (regulation status identifies the stayed route) rather than route judgement, and in this paper $w = 10$ is used, because it takes almost all of the available change-bias reduction at the smallest cost on revisions. The same run-to-run variation separates the sweep's 68.2 % from the 68.1 % headline.

7.7. What the model appears to weigh

A ranking model that cannot be interrogated is of little use to an operational user who is asked to trust it, and Shapley attributions were therefore computed over the held-out quarter. Principles from game theory can be used to interpret the prediction of a model for a given input, assuming that each feature is a player and that the model output is the payout. Let us consider the following scenario: all the features participate in the game (i.e., contribute to the model output) and enter the room where the game is played in a random order. The attribution of a feature is the average change in the payout received by the coalition already in the room when the corresponding player joins them, and this measure is the Shapley value (Lundberg and Lee, 2017). Averaging their magnitude over the quarter gives Fig. 5, which is a ranking not of what correlates with acceptance, but of what moves the score of this particular model.

Two results stand out. First, the waypoint sequence is the single largest term, ahead of every cost indicator, which means that route structure carries information about the choice that the four cost indicators do not, and which is precisely what a cost-only model cannot represent and a learned one can. Secondly, attributed ATFM delay moves the score more than planned fuel does, which is consistent with delay being directly monetisable at planning time whereas the fuel difference between two candidate routes is often small

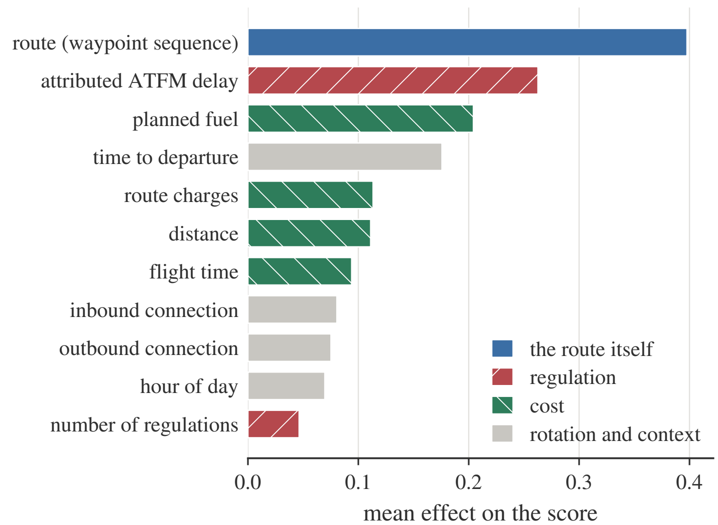

Figure 5: Mean absolute Shapley value per feature on the held-out quarter. The waypoint sequence is the largest single term, ahead of every cost indicator, and regulation exposure outweighs planned fuel.

beside it. This is a property of the score of the model rather than a statement about how any operator reasons.

Descending to individual waypoints sharpens the picture. Deleting a single identifier from a route and rescoring shifts the score by up to 2.5 points for the most influential one, and the affected waypoints cluster in a few places along the track rather than spreading evenly, which indicates that what the model has acquired is route-specific structure rather than a general preference for shorter routes. These are statistical regularities in filing behaviour across the network; they carry no assessment of any airspace, and individual waypoints and volumes are deliberately not reported.

One caveat should nevertheless be carried forward. The waypoint text ranks first by attribution, yet removing it and retraining costs only 1.4 points of accuracy. Attribution measures how much a feature moves the score of the model we have; ablation measures how much worse a model without it would be. Where features are correlated, as route geometry and cost indicators plainly are, the two answer different questions and the second is the one that governs what may safely be dropped.

Table 6: Operating points. The threshold τ is selected on the tuning month for each precision target and evaluated unchanged on the held-out quarter; fuel is planned fuel confirmed by the airlines' own subsequent filings, per quarter. CO_2 uses 3.16 kg per kg of fuel.

target [%]	τ	precision [%]	proposals	coverage [%]	fuel [kt]	CO_2 [kt]
60	0.500	65.8	98 783	35.8	24.3	76.9
70	0.525	74.1	74 201	26.9	22.7	71.6
80	0.600	79.2	56 162	20.4	20.1	63.5
90	0.800	88.2	28 756	10.4	13.8	43.6

8. The proposal channel

How much fuel is at stake, and how confident can one be that it is real? The simulation of this section answers both questions by refusing to count anything that the airline did not subsequently do.

8.1. The policy and its operating points

The policy is the following: for a flight whose candidate set contains an alternative burning less planned fuel than the filed route, propose that alternative when its calibrated acceptance probability exceeds a threshold τ, with τ selected on the tuning month for a target precision and reported unchanged on the evaluation months. In this simulation the candidate set is the revision pair itself, so the alternative considered for each flight is the other route of its own pair, and the channel is accordingly evaluated only on flights that did revise, which is the boundary Section 10 returns to. A proposal is counted as realised when the airline's own subsequent filing chose that alternative. Flights with such a cheaper alternative are the eligible flights of the tables that follow. For instance, at 80 % precision four proposals in five are confirmed in this way, and the fifth is simply not taken.

Table 6 reports the four operating points, and Fig. 6 the continuous trade behind them. Precision is bought with volume: moving the target from 60 to 90 % cuts the coverage to less than a third while giving up somewhat over two fifths of the confirmed fuel, because the proposals that survive the higher threshold are disproportionately the valuable ones. At the 80 % point used throughout this paper, the channel surfaces 56 162 proposals over three months carrying 20.1 kt of planned fuel that the airlines' own subsequent

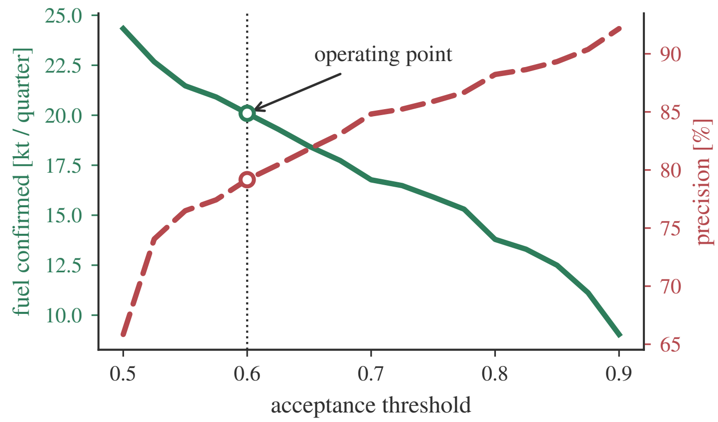

Figure 6: What the channel would deliver, as a function of how selective it is. Raising the acceptance threshold buys precision and costs volume, and the fuel that airlines subsequently confirmed falls with it. The operating point used throughout this paper is marked; it was fixed on the tuning month.

filings confirm, equivalent to 63.5 kt of CO_2. Annualised, this is 70 to 80 kt of fuel, or of order 220 to 250 kt of CO_2, where the range spans the two extrapolation bases available (a simple factor of four on the quarter, and the per-eligible-pair rate applied to the annual eligible volume estimated from the eighteen-month archive); the corresponding ranges at the other operating points follow the same pattern and reach 85 to 97 kt of fuel at the 60 % target. The median accepted proposal saves 130 kg and the distribution is strongly skewed: the top tenth of accepted proposals carries 64 % of the value.

8.2. Where the value sits

Of the 882 operators that receive any proposal, 6 fly half of the confirmed fuel and 28 fly four fifths of it (Fig. 7). By city pair the concentration is weaker but still pronounced: 153 of the 10 939 pairs carry half the value. The quarter is not carried by one month either; the three test months contribute 5.9 to 7.5 kt each. A channel can therefore start with a handful of operators and still address most of the benefit, which matters because any first deployment would be a trial rather than a rollout. One could argue that beginning with those few operators is inequitable towards the rest. Nevertheless, the concentration follows that of traffic itself rather than any difference in operator practice, and nothing in the policy withholds a proposal from any operator that qualifies for one.

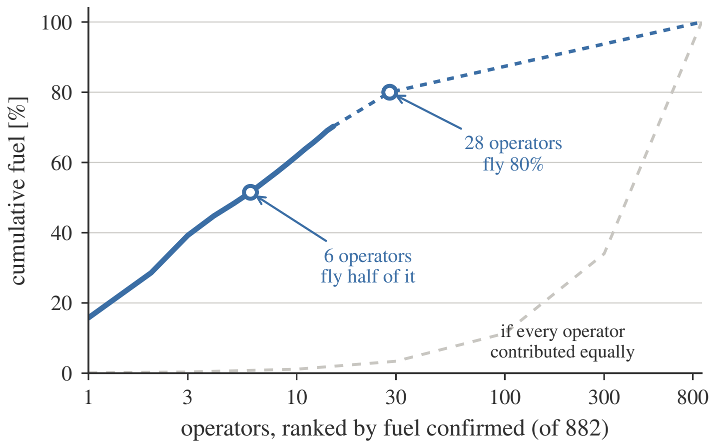

Figure 7: Cumulative share of confirmed fuel against operator rank (log axis). Six operators fly half of the value and twenty-eight fly four fifths; the dashed line is the equal-contribution reference.

8.3. Sensitivity to the bias correction

Every fuel figure above is computed with the stay-pair correction in place, and a fair question is how much of the result it moves. The whole simulation was therefore repeated with the uncorrected model. At the same threshold of 0.500 the uncorrected variant reaches 64.4 % precision against the corrected model's 65.8 %, and confirms 24.38 kt against 24.33 kt. The precision difference runs in the corrected model's favour and the fuel difference is a twentieth of a kilotonne: on flights that did revise, the correction costs nothing on this simulation metric (its one-point cost on revision accuracy, Section 7.6, is a different measure). The correction is not there to improve the simulation; it is there for the deployed population, in which most flights stay, and the honest summary is that the headline fuel figures are robust to it while the deployability argument depends on it.

8.4. A proposal, concretely

One case from the held-out quarter shows what the channel is for, and is worth following from end to end, because every step is a step that the deployed system would take.

A narrow-body was filed on a medium-haul sector, with no regulation anywhere in the picture. First, the system considers the filed route and one alternative. It then computes their indicators and subtracts the pair minimum from each, so that what reaches the model is a difference rather than an absolute quantity, and finds that the alternative burns 2 016 kg less planned fuel, about a fifth of the trip fuel, at essentially the same flight time. Both routes carry the same flight-level context, namely the same operator, the same aircraft, the same departure hour and the same inbound and outbound connections. Needless to say, nothing that the model sees distinguishes the two routes except where they go and what they cost.

The model then scores each route, and the alternative comes out +3.7 ahead. That margin is not yet a decision: it passes through the calibrator fitted on the tuning month, which converts it into an acceptance probability, and that probability is finally compared with the threshold $\tau = 0.600$ that was fixed before the evaluation period began. This case clears the threshold, and the policy therefore proposes. Had the margin been smaller, the flight would simply have been left alone: the system is permitted to say nothing, and on roughly four flights in five that is what it does.

The operator's own later filing chose exactly that route. Keep in mind, however, that the airline was told nothing, so this is evidence not that a proposal would have been accepted, but that the model identified, before departure, a route the airline itself went on to prefer. That is the strongest validation available without running the channel, and the form on which every fuel figure in this paper rests.

9. How fast does the model age?

A model that scores well on average but decays from month to month is not deployable, and the main split can only partly address the question: its three test months rule out rapid decay, but they say nothing about the retraining cadence a deployed system would need. This section answers with a second experiment. The entire design of Section 4.4 is moved back one year: the same configuration (stay weight 10, within-pair normalisation, waypoint text) is fitted on January to October 2025, tuned and calibrated on November and December 2025, and then evaluated month by month on January through June 2026, so that test month m sits m months beyond the training cutoff. Nothing from 2026 touches the model, the calibrator or the threshold, which is selected on the tuning months for the 80 % precision target exactly as in Section 8.

Training proceeded exactly as for the production configuration (10 000 iterations at the automatically selected learning rate, early stopping at iteration 9 386), and the threshold selected on the 2025 tuning months for the

Table 7: The year-2025 model evaluated month by month across the first half of 2026 (Wilson 95 % intervals). Coverage is the share of eligible flights on which the channel proposes at the threshold fixed on the 2025 tuning months; the regulated share is the fraction of test pairs whose newly filed route carried attributed delay.

month	months past cutoff	accuracy [%]	regulated share [%]	coverage [%]	precision [%]
January	1	67.2 [66.8, 67.5]	5.7	22.2	78.9 [78.3, 79.6]
February	2	66.1 [65.7, 66.5]	5.1	21.3	78.6 [77.9, 79.3]
March	3	64.6 [64.2, 65.0]	5.2	20.4	78.0 [77.3, 78.7]
April	4	65.6 [65.2, 65.9]	8.6	19.8	78.8 [78.2, 79.5]
May	5	66.4 [66.1, 66.7]	16.2	18.7	77.2 [76.6, 77.9]
June	6	68.0 [67.7, 68.3]	22.6	17.3	77.8 [77.2, 78.5]

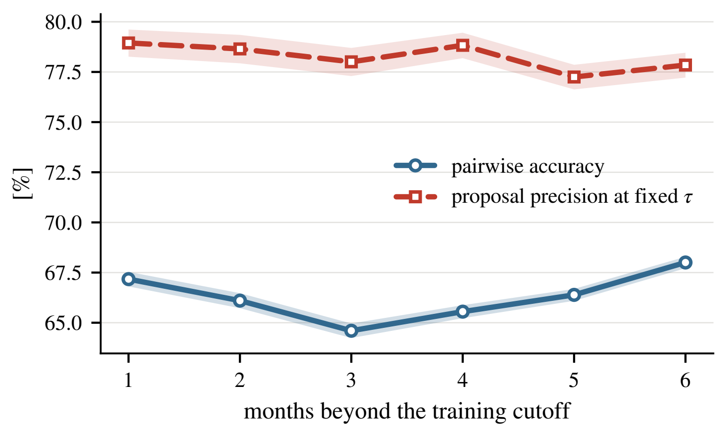

Figure 8: How the model ages. Pairwise accuracy (solid) dips and recovers across the half year, while proposal precision at the year-old threshold (dashed) never leaves a band of under two percentage points. What declines steadily is coverage (Table 7): the ageing model proposes less, and errs no more.

80 % precision target came out at $\tau = 0.600$, the same value that the main split produced a year of data later. Table 7 reports the six months in full, and Fig. 8 the two curves that matter.

Pairwise accuracy starts at 67.2 % one month beyond the cutoff, drifts down to 64.6 % at three months, and then recovers to 68.0 % at six. Interestingly, the worst month is not the furthest one: the curve is U-shaped, and the best of the six months is the last. Two mechanisms plausibly contribute. First, the mix shifts: the regulated share of the test pairs grows from 5.7 %

in January to 22.6 % in June as the summer regulation season arrives, and Section 7.3 showed that regulated pairs are the easier stratum. The shift does not explain the recovery on its own, however, because the same U-shape appears within the unregulated stratum alone (67.0 % down to 64.7 % and back to 67.0 %). Secondly, we hypothesise a seasonal re-alignment: June 2026 resembles the June 2025 that sits inside the training window more than the late-winter months resemble anything the window emphasises. Elapsed time and season are confounded by construction in any single such experiment, and the data cannot separate them cleanly; what the experiment does establish is that six months of staleness never cost more than a few points. Where the two designs overlap the comparison is direct: a model trained through February 2026 scores 67.5 %, 67.7 % and 69.0 % on April, May and June, so the year-2025 model gives up 1.9, 1.3 and 1.0 points on the same months, and 2.9 points on March against the fresher model's score on its own tuning month. The accuracy cost of roughly a year of staleness is therefore one to three points, largest at the seasonal antipode.

The operational question is what happens to the channel, and the answer is the striking part. At the threshold fixed on the 2025 tuning months, precision on proposals is 78.9 % one month out and 77.8 % six months out, and never leaves the band from 77.2 to 78.9 % across the half year; the main split's operating point, tuned a year later, delivered 79.2 % on its own quarter, and the aged model sits about a point below it on this measure. What moves instead is coverage, which falls from 22.2 % of eligible flights in January to 17.3 % in June, about a fifth in relative terms. As the traffic drifts away from what the model was trained on, its calibrated confidence clears the threshold less often, and the channel becomes quieter rather than wrong. For a proposal channel this is the correct failure mode, because a wrong proposal costs trust in the way Section 1 describes, whereas a missed one merely leaves fuel where it was. It should be noted that the monthly eligible volume nearly doubles across the window as summer traffic builds, so part of the coverage movement within the half year is seasonal composition rather than ageing. On the held-out quarter, however, the two designs face the same flights and the season cancels exactly: over April to June the aged model proposes on 18.5 % of the eligible pairs against the fresh model's 20.4 %, at 77.9 against 79.2 % precision. The net cost of roughly a year of staleness to the channel is therefore about a tenth of the proposal volume and a point of precision, which is the cleanest isolation of pure ageing this data affords.

For a deployment plan the reading is straightforward. Retraining quarterly would recover the one to three accuracy points that staleness costs, and even an annual cadence would hold proposal precision within two points of the fresh model's at the price of roughly a fifth of the proposal volume. Nevertheless, six months and one seasonal transition are the extent of the evidence, and nothing here licenses extrapolation to a model several years old, still less to one serving traffic whose filing behaviour a live channel has itself begun to influence.

10. Threats to validity

The limitations of this study are collected here rather than scattered, in the order of how much they bound the claims.

The label is behaviour, not utility. A revision records what the operator's filing process produced, which folds together the airline's economics, its dispatching software, its habits and its workload. The model learns this compound, and the compound is arguably the right target for a proposal channel, since a proposal is accepted or ignored by the same compound. It does mean, however, that nothing in this paper measures a preference in the economic sense of Samuelson (1938) separately from the process that expresses it, and transfers of the learned function to settings where the process differs (e.g., another region's filing culture) should be expected to degrade.

Flights that never revise. Every revision pair is a flight that changed its plan. A deployed channel would see every flight, and most flights never revise, so the model would be asked a question on which it was not evaluated. The honest expectation is that precision falls, because the flights that never revise are precisely the ones whose operators were content with what they filed. Section 7.6 quantifies how much of that gap the stay pairs close: some of it, not enough. The stay construction itself is confounded by design, since regulation status identifies the label, and the masked evaluation is a repair rather than a clean design. A balanced stay dataset in which regulation status is independent of the label requires data that is not currently assembled rather than data that does not exist, and is the single most valuable extension of this work.

Planned fuel is not burned fuel. Every fuel figure rests on planned fuel, and the two differ by whatever the tactical phase imposes (e.g., a direct routing granted en route, or a cruise level capped below the one requested). Quantifying the difference requires flown trajectories matched to the plans that preceded them, and until that is done the totals of Section 8 should be read as an upper bound on what a proposal channel would deliver.

The counterfactual is absent. On every flight the simulation counts, the operator found the cheaper route by itself, so the 20.1 kt is not fuel that a channel would create but fuel that a channel could have surfaced earlier. What a channel would genuinely add, namely the flights on which the alternative was never considered, is precisely what this data cannot measure. Only a trial in which real proposals are made, and their acceptance logged, can close this gap, and such a trial is also the only way to observe acceptance itself rather than its revealed proxy.

Data currency at serving time. Attributed delay and the regulation picture are among the strongest inputs, and both are properties of a moment rather than of a flight. A score computed against a regulation state that has since changed is a score answering a question that nobody asked, and the binding constraint at serving time is therefore data currency rather than computation: the scoring step is a single pass through a fixed tree ensemble.

11. Conclusions

Airline route acceptance is learnable from flight-plan revisions at network scale, and the findings of this paper are fivefold. First, a model fitted on 1 134 000 revisions identifies the chosen route in 68.1 % of held-out cases and supports operation at a selected precision rather than at a single accuracy figure: on the most confident fifth of the decisions it is right 94.9 % of the time. Secondly, every rule of the form "prefer the cheaper route" scores below chance on this archive, and the model's advantage concentrates exactly where cost logic fails: 79.9 % on moves against all three cost indicators, against 53.9 % on moves the indicators already favoured. Thirdly, a proposal channel operating at 80 % precision addresses 20.1 kt of planned fuel per quarter, validated against what the airlines subsequently filed, four fifths of it flown by 28 of 882 operators, and these figures are robust to the stay-pair bias correction. Fourthly, the model weighs route structure more heavily than any individual cost indicator, which is precisely the information that a formulation based on costs alone cannot represent. Fifthly, trained on one year and watched over the following six months, the model's proposal precision never leaves the band from 77.2 to 78.9 %, somewhat below the 80 % target it was tuned for, while its accuracy dips by at most three points before recovering with the season: it ages by proposing less, not by erring more.

One practical note should be made before turning to the limits. The number that characterises this system is not 68.1 % but the curve of Fig. 4, because a channel chooses where on that curve to sit. Note that ranking a pair correctly and proposing correctly are different quantities, and that the second of them is the operational one: at the threshold used here, the channel proposes on about a fifth of the eligible flights, and 79.2 % of those proposals match what the airline went on to file. Accordingly, any first deployment would sit well to the left of that curve, and would move right only as acceptance is observed.

Future work should be directed, in the first place, at assembling a balanced stay dataset in which regulation status is independent of the label, which is what currently bounds the change-bias measurement; in the second place, at matching flown trajectories to the plans that preceded them, so that planned savings can be restated as burned ones; and in the third place, at a shadow-mode trial logging the real responses of operators, which is the only way to observe acceptance rather than its revealed proxy. Last but not least, the horizon experiment of Section 9 gives a deployment plan its retraining cadence, but only a live channel can reveal how preferences shift once proposals themselves begin to influence filing behaviour, and any deployment should therefore expect to re-measure everything this paper measures.

Disclaimer

The views expressed are those of the authors and do not necessarily reflect the official position of EUROCONTROL or of its Member States. This paper reports research and does not commit to any operational service.

CRediT authorship contribution statement

Ramon Dalmau: Conceptualization, Methodology, Software, Data curation, Formal analysis, Investigation, Visualization, Writing – original draft. **David Perez:** Conceptualization, Resources, Validation, Writing – review & editing. **Seddik Belkoura:** Methodology, Validation, Writing – review & editing. **Vincent Taverniers:** Resources, Data curation, Writing – review

& editing. **Robin Deransy:** Conceptualization, Project administration, Writing – review & editing. **Benjamin Cramet:** Supervision, Writing – review & editing. **Gregory Gustin:** Supervision, Project administration, Writing – review & editing.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Claude (Anthropic) to assist with drafting and editing the manuscript text and with producing the analysis figures from the study's result tables, and Gemini (Google) to develop the layout of the methodology schematic of Fig. 2, which was then redrawn as vector code. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data used in this study is operational air traffic data held by EUROCONTROL and processed under existing data-handling arrangements; it cannot be made publicly available. Aggregated tables underlying every figure are available from the corresponding author upon reasonable request.

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